25	**D**	The oil droplet is stationary, so the electric force balances its weight. $$qE = mg$$ For the second droplet: • Charge $= \frac{q}{2}$ • Radius $= 2r$ Since it is the same oil, its density is unchanged. Mass is proportional to volume: $$m \propto r^3$$ Hence, $$m_2 = (2^3)m = 8m$$ The new equilibrium condition is $$\frac{q}{2}E_2 = 8mg$$ Using the original condition $mg = qE$, $$\frac{q}{2}E_2 = 8(qE)$$ Cancel q: $$\frac{E_2}{2} = 8E$$ $$E_2 = 16E$$
26	**C**	
27	**B**	For a straight current-carrying wire in a magnetic field